



Personnel Warehouses Auxiliary services

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Ch4 : Personnel requirements include

- Parking
- Entrance/exit
- Lockers for storage of personal belongings
- Restrooms
- Food services
- Health/medical services
- Recreational services
- Offices (also SMART offices)

*Difficult to plan due as it is dependent on management philosophies
(see examples in book on how it can differ)*

Ch7: Warehouse operations

- Sometimes it is difficult to understand whose responsibility it is...operations? Logistics? Operation excellence? Or even IT?
- Sometimes a company's core business = warehousing (Take-a-Lot, DHL, DSV)
- Large warehouses often include RECEIVING and SHIPPING functions (p392)
- Some principles...7.4.1
- (NB – stuff must be stored such that it can be FOUND)
- Steps to determine SPACE requirements:
 - *Step1 Determine WHAT, HOW MUCH and WHEN stuff is to be received and shipped (fig 7.4)*
(include the idea of STAGING)
 - *Step2 Determine the NUMBER and TYPE of 'docks' (waiting line analyses)*
 - *Step3 Determine space for In/Out WITHIN facility (remember the 'holding' area as well as the need for maneuvering of transport)*

Ch7: Warehouse operations

- Docks – locations, operations, levelers
- Ex 7.1 – note all the detail specs that determines the calc
- Storage space planning (influenced by Inventory management policies)
- Ex 7.2 as it relates to Inventory management decisions
- Ex 7.4 ABC layout idea demonstrated (A most moving vs C least moving items)
- Storage layout planning according to principles (p424 NB)

Popularity/ Similarity/ Size/ Characteristics (eg perishable, oddly shaped, crushable, hazardous, security, compatibility eg chemicals)

- Fig 7.28 Proper and improper layout examples
- Concept of a unit load = NB design determining parameter
- Use 3D space as there is less movement
- Determination of aisles = key factor in warehouse design

Ch9: Facility auxiliary systems

- These are often architects, civil engineers, builders responsibilities
- BUT the facility planner need to know WHAT systems are required WHERE
- Systems to include:
 - Structural (walls, columns, roofs)
 - Enclosure (barriers against wind/water/humans incl: walls, windows, floors and for comfort incl: doors, ventilation, thermal)
 - Atmospheric (temperature incl heating/cooling, humidity, clothing, “clean” rooms)
 - Electrical (lighting and other systems requiring electricity)
 - Life safety (fire protection, seismic events and power failures)
 - Sanitation (water, plumbing)
- The idea of Automation (think SMART)
- Sometimes Maintenance functions are also seen as part of auxiliary systems